INCH-POUND

MIL-G-29635

20 November 1990

MILITARY SPECIFICATION

GLOVES, FIREMEN'S, STRUCTURAL/ALUMINIZED PROXIMITY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for a flame resistant combination structural/aluminized proximity firemen's glove.

1.2 Classification. The gloves shall be of one type only and in the following sizes (see 6.2).

X-Small (XS)
Small (S)
Medium (M)
Large (L)
X-Large (XL)

2. APPLICABLE DOCUMENTS

2.1 Government documents.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Officer in Charge, Navy Clothing and Textile Research Facility, P.O. Box 59, Natick, MA 01760-0001 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8415

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2.1.1 Specifications, standards, and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

KK-L-2004	- Leather, Cattlehide, Deerskin and Horsehide, Chrome Tanned
NN-P-71	- Pallet, Material Handling, Wood, Stringer Construction, 2-Way and 4-Way (Partial)
DDD-L-20	- Label; for Clothing, Equipage and Tentage (General Use)

MILITARY

MIL-C-3735	- Cuffs, Knit, Wrist and Ankle, and Cloth, Knitted
MIL-P-15011	- Pallets, Material Handling, Wood, Post Construction, 4-Way Entry
MIL-B-17757	- Box, Shipping, Fiberboard (Modular Sizes)
MIL-C-21852	- Cloth, Taffeta, Nylon
MIL-C-24929	- Cloth, Laminated, Aluminized (Para-aramid base)
MIL-C-24941	- Cloth, Laminated, Aluminized (Para-aramid/PBI base)
MIL-T-44100	- Thread, Para-aramid, Spun, Intermediate, Modulus
MIL-C-87059	- Cloth, Knitted, Nylon, Tricot

STANDARDS

FEDERAL

FED-STD-191	- Textile Test Methods
FED-STD-311	- Leather, Methods of Sampling and Testing
FED-STD-751	- Stitches, Seams, and Stitchings

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	- Marking for Shipment and Storage
MIL-STD-147	- Palletized Unit Loads
MIL-STD-2073-1	- DOD Materials Procedures for Development and Application of Packaging Requirements
MIL-STD-2073-2	- Packaging Requirement Codes

(Unless otherwise indicated, copies of federal and military specification, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

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2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

U.S. POSTAL MANUAL

(Copies of the manual may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402-0001.)

2.2 Non-Government publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issue of documents which are DoD adopted shall be those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification

(Applications for copies should be addressed to the American Trucking Association, ATTN: Traffic Department, 1616 P Street, N.W., Washington, DC 20036-1404.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606-5808.)

NATIONAL FIRE PROTECTION ASSOCIATION

NFPA 1973 - Gloves for Structural Firefighting

(Applications for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269.)

AMERCIAN SOCIETY FOR TESTING AND MATERIALS

ASTM E-96 - Water Vapor Transmission of Materials
ASTM D1424 - Tear Resistance of Woven Fabrics by Falling Pendulum (Elmendorf) Apparatus

(Applications for copies should be addressed to the Amercian Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

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2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Guide sample. Samples, when furnished, are solely for guidance and information to the contractor (see 6.3). Variations from this specification may appear in the sample in which case this specification shall govern.

3.2 First article. When specified, the contractor shall furnish sample unit(s) for first article inspection and approval (see 4.3 and 6.2).

3.3 Material.

3.3.1 Leather. The palm area shall be a blue/gray chrome tanned flesh split cattlehide conforming to type VIII of KK-L-2004. The leather shall be a 3 to 3-1/2 ounce thickness after silicone treatment specified in 3.3.1.1 except that:

a. Shrinkage temperature shall not be less than 105°C.
b. Test for percent ash and chloroform soluble material are not required.

3.3.1.1 Silicone treatment. The leather shall be treated with an approved durable silicone compound (see 6.4) to lower its water absorption. The treated leather shall not absorb more than 30 percent water when tested as specified in 4.4.1.

3.3.2 Aluminized shell. The aluminized shell for the back of the glove shall be either of the following fabrics: A highly reflective aluminum surface laminated to a knitted para-aramid base fabric conforming to Type II of MIL-C-24929 or a highly reflective aluminum surface laminated to a knitted para-aramid/PBI base fabric conforming to MIL-C-24941.

3.3.3 Liner.

3.3.3.1 Liner material. The material used in the glove lining shall be a 40/35/25 (± 2.5%) blend of para-aramid, PFR rayon and PBI (polybenzimidazole). The liner material shall be made by using a 3 bar rashel knitting machine. The front and back bars shall be 32 singles consisting of 40/60 PBI/para-aramid; and the middle bar shall be 15 singles consisting of PFR rayon. The technical back shall be napped (the side of the fabric that the rayon yarns are predominant). The use of fibers other than the approved is prohibited. The cloth shall conform to the physical requirements listed above and in Table I when tested as specified in 4.4.1.

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Table I - Physical Requirements

Characteristics	Minimum	Maximum
Wales per inch	20	
Courses per inch	20	
Tear strength - lbs (N)		
Wale	10 (44.5)	
Course	8 (35.6)	
Weight - oz/yd^2 (g/m^2)	8.0 (271.2)	
Bursting strength - lbs (N)	90 (400.5)	
Flame resistance, vertical (Wale direction only)		
After flame time, seconds		0.5
Char length, inches (cm)		1 (2.54)
After glow, seconds		15

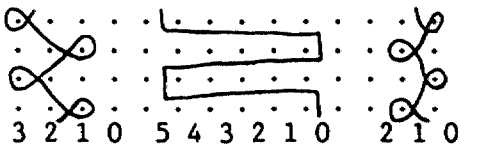
3.3.3.2 Para-aramid. The fiber shall be 1-1/2 denier, 2 inch staple, when tested as specified in 4.4.1.

3.3.3.3 Rayon. The PFR (permanently flame retardant) rayon is manufactured by metering water-insoluble phosphorous-base chemical compounds into viscose solution from which rayon fibers are spun. The fibers contain the flame retardant which is highly dispersed in a matrix of PFR cellulose. The fiber shall be a nominal 1.5 - 2.0 denier and nominal 1.5 - 2.0 inch staple cut length, when tested as specified in 4.4.1.

3.3.3.4 PBI. The fiber shall be 1-1/2 denier, 2 inch staple, when tested as specified in 4.4.1.

3.3.3.5 Color. The color shall be natural.

3.3.3.6 Knit construction. The guide bar stitch pattern shall be as follows when tested as specified in 4.4.1.



Front guide bar #1: 1-0/2-3
Middle guide bar #2: 0-0/5-5
Back guide bar #3: 1-2/1-0

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3.3.4 Insert. The glove insert shall be made of nylon tricot fabric conforming to MIL-C-87059, except the construction shall have a minimum of 36 wales and 24 courses per inch. The fabric shall have a waterproof, moisture vapor permeable barrier (membrane, coating or laminate) applied such that the barrier is orientated toward the hand in the glove. The insert and sealed seam shall meet the following requirements when tested in accordance with 4.4.1.

Characteristic	Requirement
Water penetration resistance (psi) - Minimum	4.0
Moisture vapor transmission rate (g/m^2/24 hr) - Minimum	400

3.3.5 Cuff. The fabric used for the cuff shall be in accordance with type IV, class 3 of MIL-C-3735, except the yarn shall consist of 60% para-aramid and 40% PBI (polybenzimidazole). The fabric shall meet the following requirements when tested in 4.4.1.

Characteristic	Requirement
Flame resistance	
After flame (seconds) - Maximum	2.0
After glow (seconds) - Maximum	8.0
Char length - inches (cm) - Maximim	4.0 (10.16)

3.3.6 Adhesive. The adhesive used to bond the inside of the insert fingertips to the lining shall be compatible with the materials utilized.

3.3.7 Thread. The thread for assembling the glove shall be a para-aramid spun staple thread, size T-70, minimum of 4 ply, natural color conforming to MIL-T-44100.

3.3.8 Label. Each glove shall have a combination warning, size, identification, instruction label conforming to type VI, class 14 of DDD-L-20, except that colorfastness requirements shall not apply. The label shall have the following format and inscription:

GLOVES, FIREMEN'S, STRUCTURAL/ALUMINIZED PROXIMITY
SIZE: MEDIUM OR M (EXAMPLE)
CONTRACT NO: DLA-100-00-000-0000 (EXAMPLE)
STOCK NO: 8415-00-000-0000 (EXAMPLE)
NAME OF CONTRACTOR:
NAME OF MANUFACTURER: (IF OTHER THAN CONTRACTOR)

WARNING

DO NOT KEEP IN DIRECT CONTACT WITH FLAME
KEEP CLEAN, SOILING WILL REDUCE PROTECTIVE QUALITIES
HAND WASH IN COLD WATER WITH LIQUID SOAP
DO NOT BLEACH
AIR DRY ONLY

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3.4 Design. The glove shall be five fingered, gunn cut, with a straight thumb and knit wristlet. Palm shall be leather, and back shall be aluminized fabric, with aluminized side on the outside. A waterproof moisture vapor permeable glove insert shall be sandwiched between the inseamed shell and napped knitted lining.

3.4.1 Figures. Figures are furnished for information purposes only. When inconsistancies exist between the written specification and the figure, the written document shall govern.

3.5 Patterns. Standard patterns to be used to cut working patterns will be furnished by the Government (see 6.3). The working patterns shall be identical to the Government patterns. Neither the Government patterns nor the working patterns shall be altered in any way, except that additional notches for use during construction are permitted on the working patterns. Also, minor modifications are permitted where necessary when using automatic equipment. These modifications shall not alter the dimensional, serviceability, or appearance requirements cited in the specification. The seam allowance shall be 1/8 inch.

3.5.1 Pattern parts. The component parts shall be cut from the materials as specified, and in accordance with the following list of pattern parts:

Material	Pattern parts	Cut parts
Leather shell	Palm	2
	Thumb	2
	Fingers	2
Aluminized shell	Back	2
	Wrist tab	2
Knit liner	Back liner	2
	Palm liner	2
	Thumb liner	2
	Finger liner	2

3.6 Construction.

3.6.1 Stitches, seams and stitchings. Stitch, seam and stitching types specified in Table II shall conform to FED-STD-751. Whenever two or more methods for seam or stitch types are given for the same part of the operation any one may be used. Seam allowance shall be maintained with seams sewn so that no raw edges, runoffs, twists, pleats, puckers or open seams result. All seams shall start and finish evenly. Thread tension shall be maintained so that there is no tight or loose stitching. The bight of the overedge stitching shall be 3/16 inch. The guide and knife shall be set to trim only the ravelled ends of the fabric. In order to minimize the size of the needle hole in the leather, the needle used in the seaming operations shall be of the smallest size which will accomodate the specified size of thread.

3.6.2 Stitches per inch. The minimum and maximum number of stitches per inch shall be as specified in Table II.

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3.6.3 Thread breaks and ends of seams. Ends of all seams and stitching produced with 301 stitch type, when not caught in other seams or stitching, shall be securely tacked or backstitched not less than 3/8 inch. Thread breaks (all stitch types) or bobbin run-outs shall be secured by stitching back of the break not less than 1/2 inch.

3.6.4 Cutting. The leather shall be examined on both sides prior to cutting and any area that contains surface defects shall not be included in the cut leather parts. The commercial method of pre-measuring and adjusting the stretch of the skin to provide measurements for the cut tranks, thumb, and fingers will be permitted provided the finished measurements of Table III are maintained. Skins shall not be toggled and set whereby all stretch of the leather is eliminated. Tranks and cut parts shall be matched for color, grain, and thickness of leather.

3.7 Manufacturing operations requirements. The gloves shall be manufactured in accordance with all operation requirements specified in Table II. The contractor is not required to follow the exact sequence of operations listed, provided that the gloves are identical to those produced by following the sequence of operations as listed in Table I. The length (cuff to fingertip) direction of the cloth parts shall be cut in the wale direction of the fabric. Minor modifications are permitted where necessary when using automatic equipment. These modifications shall not alter the dimensional, serviceability, or appearance requirements cited in the specification.

3.7.1 Marking. The component parts of the gloves shall be marked, ticketed, or bundled to insure a uniform shade and size throughout the pair. Any method may be used except:

a. Corrosive metal fastening devices.
b. Sew-on shade tickets or metal fastening devices on leather shell.
c. Adhesive type tickets which show discoloration, or the adhesive mass adheres to the material upon removal of tickets.

NOTE: The use of ink pad numbering machine, rubber stamp, or pencil is allowed, provided the numbering does not show on the outside of the glove and wherever possible, is covered by the seam allowance.

3.7.2 Abbreviations in table of operations. The abbreviations used in Table II are as follows:

Stch	-	Stitch
in	-	inch
Ndl	-	Needle
Bob	-	Bobbin
Lpr	-	Looper
Mchne	-	Machine
Brtck	-	Bartack
Comrcl	-	Commercial
smlr	-	similar
Btnhl	-	Buttonhole
incl	-	including
dbl	-	double
chnstch	-	chainstitch

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TABLE II - CONSTRUCTION OF GLOVES

No.	Operation	Stch Type	Seam/ Stch type	Stch in	Thread Ndl	Thread Bob/ Lpr
1.	Cutting.					
	a. Before cutting the leather, all skins shall be examined for defects which affect the appearance of the end item. The skins shall be dampened with tap water only, and the lengthwise stretch (from butt end to head end) shall be pulled out before cutting. The skins shall be stretched in the width direction, however, sufficient stretch shall remain to prevent constriction or binding across the knuckles in the finished glove when fingers are doubled under to make a fist.					
	b. The length of the glove pattern shall run parallel with the length of the hide.					
	c. Cut liner in accordance with patterns which show directional lines for cutting and notches for proper assembly. The unnapped side shall be the finished side of the liner.					
2.	Replacement of defective components.					
	a. During the cutting and manufacturing process, components having material defects or damages that are classified as defects in section 4 shall be removed from production and replaced with non-defective and properly matched components.					
3.	Marking.					
	a. Mark, ticket, or bundle all component leather cut parts to insure a uniform shade and size for proper assembly throughout the gloves (see 3.7.1).					
4.	Prepare outer shell.					
	a. With edges aligned, stitch gunn cut fingers to palm shell.	301	SSa-1	8-10	50	50
	b. With edges aligned, stitch thumb shell to palm shell.	301	SSa-1	8-10	50	50
	c. With cut edges aligned, close the shell by stitching palm with attached fingers and thumb to back, taking care to prevent twisting or distortion of fingers. Each end of the closing seam shall be secured by backstitching not less than three stitches.	301	SSa-1	8-10	50	50

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TABLE II - CONSTRUCTION OF GLOVES

No.	Operation	Stch Type	Seam/ Stch type	Stch in	Thread Ndl	Thread Bob/ Lpr
5.	Prepare liner.					
	a. With back (unnapped) sides together and edges aligned, stitch gunn cut finger liner to palm liner.	301	SSa-1	8-10	50	50
	b. With back sides together and edges aligned, stitch thumb liner to palm liner.	301	SSa-1	8-10	50	50
	c. With back sides together, close the liner by stitching the palm with attached fingers and thumb to back, taking care to prevent finger distortion.	301	SSa-1	8-10	50	50
6.	Assemble glove.					
	a. Superimpose glove insert specified in 3.3.4 onto outershell aligning respective finger tips. Stitch finger tip tab extensions of glove insert along curved seam allowance of shell fingers. To maintain the waterproof integrity of the glove no stitches shall penetrate the hand cavity of the glove insert.	304	Smlr to SSa-1	12-14	50	50
	b. Cement knit liner fingertips to back of respective glove insert fingertips allowing adequate drying time to tack. Turn shell to finished position with liner and insert smoothly extended inside. The seams shall be smooth and parallel and the finger tips shall be fully extended. Remove all wrinkles, creases, or lumps which may affect the comfort when worn.					
7.	Attach wristlet and label.					
	a. Fold and double wristlet. Stitch the wrist tab to the outside of the cuff through all plies, 1/8 inch from edge of tab, around the curved edge of the tab, simultaneously catching the label to the inside of the cuff. The straight edge of the tab and label shall be aligned with the raw edge of the cuff.	301	SSv-1	8-10	50	50
	b. Position cuff and outer shell with the aluminized tab centered on the aluminized back. Sew wristlet and label to glove opening. The seam allowance shall be 1/2 inch when attaching wristlet to glove.	301	SSa-1	8-10	50	50
	c. Overedge glove opening catching edge of label. Turn wristlet down.	506	SSa-1	8-10	50	50

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TABLE II - CONSTRUCTION OF GLOVES

No.	Operation	Stch Type	Seam/ Stch type	Stch in	Thread Ndl Bob/ Lpr
8.	Pairing. a. The glove shall be matched, paired, placed palm to palm and tacked. Tacking shall be done with not less than two stitches, using a suitable ply cotton thread which will permit the gloves to be separated without tearing the wristlet.				
9.	Clean gloves. a. Trim all thread ends inside and out. b. Remove size tickets or markings. c. Remove all spots and stains.				

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3.8 Measurements. The finished gloves shall conform to the dimensions specified in Table III. All measurements and tolerances are expressed in inches. All measurements shall be taken without distorting the finished glove.

Table III - Finished measurements

Size	Overall Length (A)	Width (B)	Palm Length (C)	Thumb Length (D)	Cuff Length (E)
X-Small	12-3/4	4-1/4	8-3/4	2-3/4	4-1/4
Small	13	4-1/2	9	3	4-1/4
Medium	13-1/4	4-3/4	9-1/4	3-1/8	4-1/4
Large	13-1/2	5	9-1/2	3-1/4	4-1/4
X-Large	13-3/4	5-1/4	9-3/4	3-3/8	4-1/4
Tolerances:	± 1/2	± 1/4	± 3/8	± 1/4	± 1/4

NOTE: Refer to Figure 1 for A, B, C, D, and E measurements.

(A) Measurement shall be taken from tip of second finger to top edge of knitted cuff.

(B) Measurement shall be taken across palm at thumb crotch, folded edge to folded edge.

(C) Measurement shall be taken from the wrist seam of the palm to tip of second finger.

(D) Measurement shall be taken from tip of thumb to thumb crotch.

(E) Measurement shall be taken from top edge of cuff to wrist seam.

3.9 Workmanship. After completion of the final assembly, the gloves shall be thoroughly cleaned and all thread scraps, lint, and foreign matter shall be removed. The gloves shall not contain any spot, stain, cut, abraded area, tear, mend, needle chew, or ragged edge. The gloves shall conform to the quality of product established by this document and the occurence of defects shall not exceed the acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

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4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the government to acceptance of defective material.

4.1.2 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

1. First article inspection (see 4.3).
2. Quality conformance inspection (see 4.4).

4.3 First article inspection. When required, the first article submitted in accordance with 3.2, shall be inspected as specified in 4.4.3 for compliance with design, construction, workmanship, appearance and dimensional requirements.

4.4 Quality conformance inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated.

4.4.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable procurement documents. In addition to testing provisions combined in subsidiary specifications, drawings or standards, testing shall be performed on components listed in Table IV for characteristics noted. Unless otherwise indicated, testing shall be conducted in accordance with FED-STD-191. All test reports shall contain the individual values used in expressing the final result. The leather sampling procedure and location from which the sample unit is to be obtained shall be in accordance with FED-STD-311. The lot shall be unacceptable if one or more sample units fail to meet any of the test requirements specified.

The basis for lot size and sample size for testing shall be as follows:

Component	Lot Size Expression	Sample Unit
Insert	each	Pair
Knit liner fabric	yards	1 yard
Leather	square feet	15x12x12 inch

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The lot size and sample size shall be in accordance with the following:

Lot size (yards)	Sample size
800 or less	2
801 to 22,000 inclusive	3
22,001 and over	5

Table IV - Component testing

Component	Characteristic	Requirement Paragraph	Test Method
Leather	Thickness	3.3.1	1101 1/
	Shrinkage temperature	3.3.1	7011 1/
	Silicone identification	3.3.1.1	2/
	Water absorption	3.3.1.1	8111.1 1/3/
Knit liner	Yarn blend	3.3.3.1	2/
	Weight	3.3.3.1	5041
	Yarn count	3.3.3.1	4052
	Wales/Courses	3.3.3.1	5070
	Bursting strength	3.3.3.1	5120 4/
	Tear strength	3.3.3.1	ASTM D1424
	Flame resistance	3.3.3.1	5903
	Fiber		
	Para-aramid		
	Identification, staple, denier	3.3.3.2	2/
	PFR rayon		
	Identification, staple, denier	3.3.3.3	2/
	PBI		
	Identification, staple, denier	3.3.3.4	2/
	Knit	3.3.3.6	Visual
Insert	Water penetration resistance	3.3.4	3-6 5/
	Moisture vapor transmission rate	3.3.4	ASTM E96 6/
Knit cuff	Fiber content	3.3.5	2/
	Flame resistance	3.3.5	3-3 7/

1/ FED-STD-311

2/ Unless otherwise specified, a certificate of compliance is required and will be accepted for the stated requirement.

3/ The leather shall be immersed for 30 minutes instead of the specified 24 hours.

4/ Test shall be performed with the nap side up.

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5/ Water permeability shall be measured in accordance with paragraph 3-6 of National Fire Protection Association Standard 1973, except that the samples shall be cut from the palm of glove inserts only, and shall include a seam for the full width of the specimen, and an untreated restraint cloth meeting the requirements of MIL-C-21852 shall be placed behind the specimen in the holder to minimize stretching.

6/ Five specimens shall be tested. The inner face of the insert shall face the water. The air velocity shall be 550 ± 50 feet/minute. The test chamber shall be held at 73.4 ± 2°F and 50 ± 2% relative humidity. The test duration shall be 24 hours and the samples shall be weighed at start and finish only. At the start of the test the air gap between water and specimen shall be 3/4 ± 1/16 inch.

7/ Flame resistance shall be measured in accordance with paragraph 3-3 of National Fire Protection Association Standard 1973, except that the sample shall consist of a single layer of material.

4.4.2 In-process inspection. Inspection shall be made at any point during any phase of manufacturing process to determine whether operations or assemblies are performed as specified. The Government reserves the right to exclude from consideration for acceptance any material or service in which in-process inspection has indicated non-conformance.

4.4.2.1 In-process examination of liner/insert assembly. The glove liner and insert assembly shall be examined after the completion of operation 6b of Table II. The sample unit shall be 1 glove and the selection shall be done by pairs. The lot size shall be expressed in the same terms as the sample. The inspection and acceptable quality levels (AQL's) expressed as defects per 100 units for visual examinations shall be as follows:

	AQL	Inspection level
Major	2.5	II
Total (Major and Minor combined)	6.5	II

NOTE: If one or more critical defects are found during sampling inspection, the lot shall be rejected. For each lot found to be acceptable by the sampling plan, the balance of the lot shall be inspected for critical defects only. Any critical defect found shall be removed from the lot and replaced with a conforming item.

The gloves shall be inspected for the following defects:

Defect	Classification		
	Critical	Major	Minor
Insert:			
a. Missing		X	
b. Any hole, cut, tear, mend or burn		X	

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Defect	Classification Critical	Major	Minor
Adhesive at fingertips:			
a. Missing		X	
b. Not located as specified			X
c. Incompatible, i.e., boardy, tacky, uneven, lumpy or partial delamination			X
Stitching insert fingertip to glove shell:			
a. Omitted			X
b. Penetrating seal or area inside seal	X		

4.4.3 Examination of the end item. Examination of the end item shall be in accordance with 4.4.4 and 4.4.5.

4.4.4 End item visual examination. The end items shall be examined for the defects listed in Table V. The lot size shall be expressed in units of gloves. The sample unit shall be one glove and the selection shall be by pairs. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5 for major defects and 6.5 for total (major and minor combined) defects.

Table V - End item visual defects

Defect	Classification Major	Minor
I. GENERAL		
a. Any hole (except needle hole), cut, tear, mend or burn	X	
b. Any weakening defect such as a smash, multiple float, loose slub, needle chew, or abraded area		
1. More than five:	X	
2. One to four (inclusive)		X
c. Color of any component:		
1. Not as specified		X
2. Not uniform or shaded parts clearly noticeable		X
d. Two or more thread ends not trimmed to 1/2 inch or less or two or more thread scraps not removed from the glove		X
e. Any spot or stain on outershell		X
f. Any spot or stain on liner		X
g. Any portion of the leather or cloth stiffened or scorched by any process of manufacturing	X	
h. Two or more ragged edges on the outer side of the glove		X

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Defect	Classification Major	Minor
II. LEATHER		
a. Not type specified	X	
b. Wrinkles or loose, spongy, or boardy leather:		
1. Seriously affecting serviceability	X	
2. Affecting serviceability, but nor seriously		X
c. Split vein, weak or thin spot	X	
d. Leather improperly cut, i.e., excessive stretch in length direction	X	
e. Any hole (except needlehole or pinhole) cut or tear	X	
f. Three or more needleholes or pinholes		X
III. KNIT LINER		
a. Any run, dropped stitch, snag, pull, or slubby yarn 1/		X
b. Wales not in length direction	X	
c. Stretch (stitch tension) knitting tension too tight or too loose	X	
IV. KNIT CUFF		
a. Any run, dropped stitch, snag, pull or slubby yarn	X	
b. Stretch (stitch tension) knitting tension too tight or too loose	X	
V. COMPONENTS AND ASSEMBLY		
a. Any component not as specified or any defect of assembly	X	
b. Any component, component part, or required operation omitted, or any operation improperly performed	X	
c. Back (unnapped side) of the cloth not on the exposed surface of glove interior		X
d. Insufficient stretch across the back of hand when fist is clenched causing constriction	X	
e. Fingers distorted or twisted:		
1. Seriously affecting serviceability	X	
2. Affecting serviceability, but nor seriously		X
VI. SEAMS AND STITCHING		
a. Accuracy of seaming:		
1. Any seam, hem, or attachment of any part puckered, pleated, or caught in an unrelated operation or stitching	X	
2. Any seam allowance not as specified	X	
b. Gage of stitching: 2/		
1. Any stitching irregular, unevenly gaged or not gaged as specified, or not uniform	X	
c. Any open seam 3/	X	
d. Seam and stitch type:		
1. Not specified seam or stitch		X

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Defect	Classification Major	Minor
VI. SEAMS AND STITCHING (Cont'd)		
e. Stitch tension:		
1. Any loose stitching 4/	X	
2. Any tight stitching resulting in breaking of the stitches when normal pull is applied 5/	X	
f. Stitches per inch: 6/		
1. Stitches more or less than specified:		
- by more than two stitches	X	
- by one or two stitches		X
VII. PAIRING		
a. Not right and left glove of the same size, or wide variation in the appearance	X	
b. Not tacked or joined together as specified		X
VIII. LABEL		
a. Missing, incomplete, or illegible, or size incorrect	X	
b. Improperly located		X

1/ One or two pick runs on the innerside of a glove shall not be classified.

2/ The gage of stitching defect shall be classified only when the condition exists on the major portion of the seam.

3/ A seam shall be classified as open when one or more stitches joining a seam are broken or when two or more continuous skipped or run-off stitches occur.

4/ A seam shall be classified as loose when the top and bobbin threads are on the same surface.

5/ Puckering is evidence of tight tension or gathering of the material. When puckering is evident and is not caused by the gathering of the material, the stitching shall be inspected for tight tension and breaking of stitches by exerting normal pull in the lengthwise direction of the stitching by pulling the material taut to straighten out the seam. Puckering due to gathering of material shall be classified as an accuracy of seaming defect.

6/ The stitches per inch defect shall be classified only when the condition exists on the major portion of the seam.

4.4.5 Dimensional examination of gloves. The gloves shall be examined for dimensional defects. Any dimension not meeting the specified requirement shall be classified as a defect. The lot size shall be expressed in units of gloves. The sample unit shall be one glove and the selection shall be by pairs. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 4.0. Width and length measurements shall be made while the gloves are placed on a flat surface with wrinkles and creases smoothed out. No tension shall be applied when making width and length measurements. Measurements shall be made as specified in Table III.

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4.4.6 Examination of packaging requirements. An examination shall be made to determine that packaging, packing, and marking comply with Section 5 requirements of this specification. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully prepared for delivery. Defects of closure listed below shall be examined on shipping containers fully prepared for delivery. The lot size shall be on the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 and the acceptable quality level (AQL) shall be 2.5, expressed in terms of defects per hundred units.

Examine	Defect
Marking (exterior and interior)	Omitted, incorrect, illegible, of improper size, location, sequence, or method of application.
Materials	Any component missing, damaged, or not as specified.
Workmanship	Inadequate application of components, such as: incomplete closure of container flaps, loose strapping, improper taping, inadequate stapling, bulged or distorted container.
Content	Number of items per shipping container is more or less than required. Size shown on one or more items not as specified on shipping container. 1/

1/ For this defect, one item from each shipping container in sample shall be examined.

4.4.7 Palletization examination. An examination shall be made to determine that the palletization complies with the section 5 requirements. Defects shall be scored in accordance with the list below. The sample unit shall be one palletized unit load fully packaged. The lot size shall be the number of palletized unit loads in the end item inspection lot. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 6.5 as specified in MIL-STD-105.

Examine	Defect
Finished dimensions	Length, width, or height exceeds specified maximum requirements.
Palletization	Pallet pattern not as specified. Interlocking of loads not as specified. Load not bonded with required straps as specified.
Weight	Exceeds maximum load limits.
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or C, as specified (see 6.2).

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5.1.1 Level A. Each pair of gloves shall be secured at the cuff. Six (6) pair of gloves (of one size only), with every other pair reversed shall be bundled and securely tied.

5.1.2 Level C. Glove shall be packaged to afford adequate protection against physical damage during shipment from the contractor to the first receiving authority. The package and quantity per package shall be the same as that normally used by the contractor for retail distribution.

5.2 Packing. Packing shall be Level A, B, or C, as specified (see 6.2).

5.2.1 Level A. Twelve (12) pairs of gloves (2 bundles of 6 pairs) of one size only, preserved as specified in 5.1, shall be packed in a fiberboard shipping container assembled, closed, and reinforced conforming to type CF, class weather resistant, grade 15c, variety DW, size 1A, of MIL-B-17757. The fiberboard used for the liner shall conform to type CF, class domestic, of MIL-B-17757. Each container shall have the contents completely covered on top and bottom with a sheet of commercial grade kraft paper.

5.2.2 Level B. Twelve (12) pairs of gloves (2 bundles of 6 pairs) of one size only, preserved as specified in 5.1, shall be packed in a fiberboard shipping container assembled, closed, and reinforced conforming to type CF, class domestic, variety DW, grade 200, size 1A of MIL-B-17757. The fiberboard used for the liner shall be type CF, class domestic of MIL-B-17757. Each container shall have the contents completely covered on top and bottom with a sheet of commercial grade kraft paper.

5.2.3 Level C. Items preserved, as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. The quantity per shipping container shall be the same as that normally used by the contractor for retail distribution. Containers shall comply with the US Postal Manual, Uniform Freight Classification Rules or National Motor Freight Classification Rules, as applicable.

5.3 Marking. In addition to any special marking required by the contract or order, unit packs, intermediate or exterior containers, and palletized unit loads shall be marked in accordance with MIL-STD-129.

5.4 Palletization. When specified (see 6.2), item, packed as specified in 5.2, shall be palletized on a 4-way entry pallet in accordance with load type Ia of MIL-STD-147. Each prepared load shall be bonded with primary and secondary straps in accordance with the bonding methods C and D or Film bonding F or G. Pallet patterns shall be in accordance with the appendix of MIL-STD-147. The pallet shall be 4-way, type IV; type V, class 1, size 2; or type VIII, fabricated from wood groups I, II, III or IV, grade A of NN-P-71, or 4-way, style 1, size A, type I, class 1 fabricated from groups specified of MIL-P-15011. Interlocking of loads shall be effected by reversing the pattern of each course. If the container is of a size which does not conform to any of the patterns specified in MIL-STD-147, the pallet pattern used shall first be approved by the contracting officer.

5.5 MIL-STD-2073. When specified (see 6.2), Preservation and Packing shall be in accordance with MIL-STD-2073-1 and MIL-STD-2073-2.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The gloves covered by this specification are intended to be worn by male and female personnel assigned to fight structural, shipboard, or aircraft fires.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification
b. Sizes required (see 1.2)
c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1 and 2.2)
d. When first article sample is required (see 3.2). The item will be tested and should be a first article sample. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, quality, testing and approval of the first article.
e. Levels of preservation and packing required (see 5.1 and 5.2)
f. When palletization is required (see 5.4)
g. When all packaging data is to be found in MIL-STD-2073-1 and MIL-STD-2073-2 (see 5.5).

6.3 Samples and patterns. For access to samples and patterns, address the procuring activity issuing the invitation for bids (see 3.1 and 3.5).

6.4 Silicone compounds. Only those silicone compounds approved by the U.S. Army Natick Research, Development and Engineering Center, Natick, MA 01760, and the appropriate medical service and so listed in the invitation for bids or request for proposal shall be considered acceptable for the related procurement (see 3.3.1.1).

6.5 Subject term (key word) listing.
Handwear, firefighters
Handwear, combination, firefighters

Custodian:
Navy - NU
Air Force - 99
Army - GL

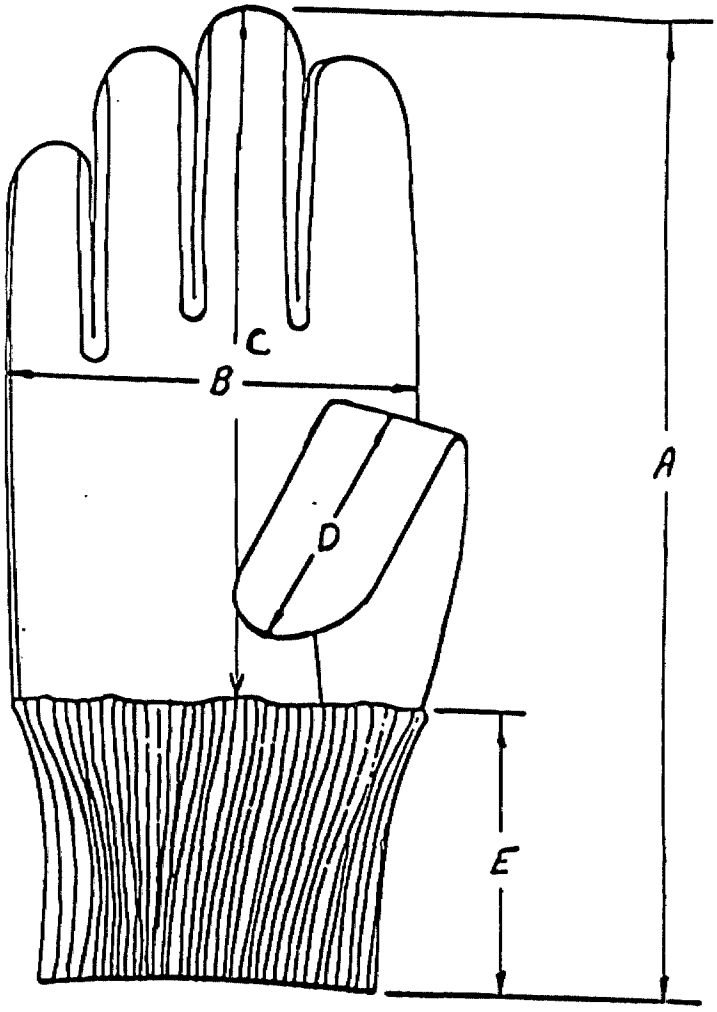
Preparing activity:
Navy - NU

Project 8415-0782

Review activities:
Navy - MC
Army - MD
Air Force - 11, 50, 82
DLA-CT

User activities:
Navy - AS, SH, CG
Air Force 45

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PALM

FIGURE 1

GLOVES, FIREMEN'S, STRUCTURAL/ALUMINIZED PROXIMITY

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